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# Guest Online Review: An Extraordinary Focus on Hotel Users' Satisfaction, Engagement, and Loyalty

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**ABSTRACT**

The article aims to heighten the impact of guest online reviews on the hotel industry. Its objective is to increase the volume of users on the degree of satisfaction for effective engagement and loyalty in the luxury hotel context. Based on a keyword-driven search, the study has examined the factors influencing the guest online reviews: guest feedback, hotel management response, and customer decision-making process. They, in turn, impact the levels of customer satisfaction and engagement in the hotel and tourism industry in Vietnam. The practical problem takes place when there is a decline in the number of customers. Particularly hotel management is involved in facing negative comments in the online market. The quantitative method tested the validity of the measurements of the questionnaire and the correlation of the constructs through SPSS tools. Three hundred and eighty-four respondents participated in the survey, who are associating with the hotel and tourism profession. The results show that the customer's decision-making process holds the most significant impact in guest reviews' online activity toward the hoteliers' service quality to create positive customer satisfaction in handling online communication. The study's implications are to foster luxury hotel management and sustainable tourism by adopting guest online reviews in developed and developing countries. Its limitation and further study are also discussed to an overall understanding of existing and future research.

**KEYWORDS**

Brand loyalty; customer engagement behavior (CEBS); customer satisfaction; online review; online travel agents (OTAS); service quality

## INTRODUCTION

Guest online reviews might help track customer satisfaction in the mass service industry, especially in hotel and tourism settings. It is increasingly becoming a vital factor in an online environment that purchasers share their comments, photos, feedback, etc., on social media sites, Online Travel Agents (OTAs), and hotel' webpage in the world of e-commerce. Therefore, to be

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successful in the present emergence of the hotel and tourism sustainable progress, almost all hotel and tourism establishments develop both their websites and co-operate with the website of OTAs. The effort is to serve these websites like sale and marketing tools in all service probabilities, contributing to the positive degree of customers' satisfaction, then influencing their engagement and loyalty (T. Vo et al., 2019a)(T. Vo et al., 2019b). The growing number of domestic and international tourists and travelers in search of guest online reviews and the inclusion of new hotels as venue' destinations for this practical purpose has raised the need for understanding the satisfaction levels and engagement behaviors of consumers toward other' opinions and feelings about the product/service of a hotel from a cross-perspective. Therefore, a striking, remarkable feature of guest online reviews is the major cause of this study to improve hotel users' volume on their satisfaction in hotel and tourism service's long-term progress.

How do the guest online reviews contribute to the efficient service performance, particularly in luxury (ranging from 4-star to 5-star) hotels in the tourism market? Experts have always seen reviews of bookers or/and online users obtain various information from former guests. Traditionally, the focus of guest reviews online has always helped potential tourists decrease perceived uncertainty. Likewise, the review readability greatly affected review helpfulness through 37,221 reviews collected from Amazon UK (Korfiatis et al., 2012) (Korfiatis et al., 2012). The review content delivered the best performance to lookers (Malik & Hussain, 2018)(Malik & Hussain, 2018). Electronic word-of-mouth (e-WOM) in social media influenced consumers' purchase intentions (Erkan & Evans, 2016)(Erkan & Evans, 2016). Accordingly, the competence for attracting international and domestic tourists has expanded to national Governments. The Viet Nam News reported that 72% of the total population use smartphones, and 48% of the portion search for hotel and travel (Anonymous, 2017a)(Anonymous, 2017a). The online booking channel in Viet Nam (VN) contributed US$671mil in 2018, including package holidays, hotel stays (volume of US$493 m), and private vacation rentals; and expected to grow 8% at US$1,918mil by 2023 (Online Travel Booking, 2018) (Online Travel Booking, 2018).

Meanwhile, the papers published also contain the following research gaps. First, the users find considerable difficulties measuring the service quality, especially in intangible products, before buying the hotel and tourism industry. The online transaction channels give travel suppliers and service providers convenient ways to engage with customers. However, most of them focus on decreasing transaction costs rather than increasing customer satisfaction and engagement through e-service quality. Customer satisfaction toward perceived service quality was likely created by guest online reviews (H. Li et al., 2013) (H. Li et al., 2013). In addition, contemporary travelers are increasingly tracking the former users' points of view when selecting accommodation. The former

customers' reviews on the online websites influenced latter consumer behaviors, mostly on OTA websites ((Ukpabi & Karjaluoto, 2016) (Ukpabi & Karjaluoto, 2016). Due to the easy-to-process information when evaluating a hotel based on certain reviews, consumers keep engaging with guest online reviews (B. A. Sparks & Browning, 2011) (B. A. Sparks & Browning, 2011). It is, therefore, the higher scores of online reviews (Lee et al., 2018) (Lee et al., 2018) or/and online rating lists of a particular hotel, the higher useful and credible especially when published by well-known online travel communities (e.g., Trip Advisor) (L. Casalóa et al., 2015) (L. Casalóa et al., 2015). Consequently, as one of OTA with a reliable source for online travel users, Trip Advisor might be measured by guest online reviews, increasing customer satisfaction and brand loyalty toward service quality. Even though this is recognized, published studies have a notable lack of research to explore this limitation.

Second, the guest reviews significantly impact online sales, with a ten percent increase in traveler review ratings boosting online bookings by more than 5% (Ye et al., 2011) (Ye et al., 2011). The combination of both messenger and message characteristics positively affects reviews' perceived usefulness (Z. Liu & Park, 2015) (Z. Liu & Park, 2015). Additionally, digital marketing had the effect of making sales volume; the higher the star rating, the higher the awarded impact (Pelsmacker et al., 2018) (Pelsmacker et al., 2018). However, some unreliable comments and reviews exist on the online platform. Accordingly, the practice of fake reviews posted online increasing customers with distrust (Ahmad & Sun, 2018) (Ahmad & Sun, 2018). In this service industry regard, the impact of e-WOM has been highlighted on customers' decision-making process (Zhang et al., 2010) (Zhang et al., 2010) and the venue's performance (Kim et al., 2016) (Kim et al., 2016). Therefore, the stream of research on e-WOM has been analyzed what factors (cause consumers to write reviews) and effects (caused by online reviews) (Cantallops & Salvi, 2014) (Cantallops & Salvi, 2014) that make customers satisfied/dissatisfied with the service by lodging managers (Mauri & Minazzi, 2013a)(Mauri & Minazzi, 2013a). Until now, few studies concern about this research area. Therefore, the researchers and practitioners need to explore more empirical study of the customer cognitive processes involved in decision-making online.

Last, another point of view is that building up a solid online reputation management strategy toward guest online reviews turns into a worthy of research direction. The roles have been considered by the frequency, speed, and response length (Vo et al., 2019a) (Vo et al., 2019b). Because too many reviews and messages on social media platforms led searchers ambivalent feelings about selecting the right reviews and making a decision toward customer engagement behaviors (Lee et al., 2018) (Lee et al., 2018). The fostering communication between hoteliers to customers (i.e., past, current, and potential) on social media has been carried on (Li, Penga et al., 2017b) (Li,

Penga et al., 2017b) to improve e-service quality customers service recovery. The number of customer satisfaction and engagement studies on the guest online reviews is limited.

Few researchers have addressed online communication, influencing customer satisfaction and engagement behavior in developing loyalty in the hotel service sector. The situation is quite severe when it prefers economic issues in a developing country like VN. This study expects to show that guest online reviews are stably and quickly becoming one of the most popular marketing and sales tools to explore customers' experience and behavior on the Internet. Since the research information-driven satisfaction from online hotel reviews has not been analyzed, customer satisfaction and engagement in the hospitality establishments have not been analyzed. Besides, previous studies have not systematically investigated the critical role of guest online reviews in the context of increasing hotel purchase probabilities and maintaining sustainable tourism in VN.

Originally, this present paper aims to seek possible ways to increase hotel guests' number toward guest reviews online in customer satisfaction. To bridge the research gaps, the findings expect to contribute a literature study on guest perception of online reviews for developing positive customers' feelings and satisfaction in the Vietnamese hotel context. We propose the significance of the hierarchical conceptual pattern of guest online review's contribution to customer satisfaction toward the high-class hotels. Especially, the study' sub-objectives are to identify dimensions influencing guest reviews on the websites as the following points: (1) *feedback of cust*omer is such one' advice to indicate one' level of experiences and satisfaction, perceived service quality at the hotel on the Internet; (2) *response of managemen*t could make the two-way communication possible, particularly when service-related complaints are expressed in online reviews by consumers; (3) and the p*rocess of decision-making* is such one believes that one' buying behavior is significantly affected by previous positive ratings from other users.

Thus, this study aims to explore the impacts of guest online reviews in practices in the hospitality and tourism industry and to answer three research questions to fill the research gaps above:

*RQ1: Are there any effects of guest online reviews (including guest feedback, management response, and customer decision-making) on customer satisfaction toward service quality in hospitality and tourism enterprises?*

*RQ2: To what extent does customer satisfaction effect customer engagement?*

*RQ3: To what extent does customer engagement effect brand loyalty?*

Taken together, these findings suggest the ethical practices of review manipulation and effective strategy of handling online interaction between users and suppliers for hotel and tourism establishments, online channel management, and online travel websites to increase levels of customer satisfaction, engagement, and loyalty toward luxury hotels on the internet.

## LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

Global trends within the online channels significantly face their beauty and drawback toward guest reviews. There are 75% of users in the world associated with social media on mobile devices (McDonald, 2018) (McDonald, 2018), and 95% of them search for hotel service quality and prices on different channels and reliable reviews before planning to reserve a room for a trip (Anonymous, 2018b) (Anonymous, 2018b). Typically, a total of 319 luxury hotels and resorts (from 4-star to 5-star hotel standard) have been built across the coastline 3,400 km long to serve high-class tourists within VN (Anonymous, 2017a) (Anonymous, 2017a). Low and medium lodging venues have accommodated while budget tourists throughout the country. Within a decade, at the subsequent contribution, total tourism receipts hit from VND30tri. to VND337tri. (from 2005 to 2015) Moreover, it modestly increased 15% growth rate over last year (Anonymous, 2017b) (Anonymous, 2017b). Following the boom of the tourism industry, despite five categories of hotel star rating (the 5-star hotel is the highest-ranking star which is developed by the Viet Nam Administration of Tourism (VNAT)), the accommodation industry in Viet Nam has suddenly risen from 3,267 hotels in 2000 to 18,800 hotels with 355,000 rooms in 2015 (Anonymous, 2017b) (Anonymous, 2017b).

In addition, there is being significantly influenced by changes in currency exchange rates of USD VND for International tourists who are coming to Vietnam absolutely using the US dollar to purchase services and products. The competition could be endless in this service sector. Consequently, travelers and tourists got benefit from this battle with lower prices, which expect to pay from US$5 for budget and US$150 for luxury options (Anonymous, 2017b) (Anonymous, 2017b). Furthermore, food and beverage services are the main components attached in accommodation service toward nutrition, food security, and good health as long as Vietnam's cuisine is one of the motivations of travelers to visit the country due to its famous with diversity, fresh and tasty ingredients. The service sector contributed almost 45% in developing countries (Anonymous, 2017b) (Anonymous, 2017b). Asian countries recently experienced international capital allocation through foreign direct investment flows toward human capital, productivity, endogenous growth, and institutional behavior (Brooks et al., 2008) (Brooks et al., 2008).

### *Service quality – an effect of guest online reviews on customer satisfaction*

The assessment and effect of service quality on the hotel's customer satisfaction have been studied by numerous scholars worldwide, such as studying the hotels in Albanian coastal (Tabaku & Cerri, 2016) (Tabaku & Cerri, 2016). Scholars have argued that guests' satisfaction measurement became a familiar practice in the United Kingdom, but with careless adequate assess especially in

poor hotel achievement (Gilbert & Horsnell, 1998) (Gilbert & Horsnell, 1998). The current study results support the argument that all those factors play a crucial role in Asia's recent growth in different degrees in different countries, with complex relationships to investment incentives. Few attempts from academic researchers have studied customer satisfaction toward service quality, increasing brand loyalty of online hotel bookers. Customers would come back if they were satisfied with the guest's online reviews (Cheung & Thadani, 2012) (Cheung & Thadani, 2012). Otherwise, they would not come back in the future (Choi, 2001). Especially, this study intends to explore the influence of guest feedback on the various internet sites toward repurchasing perceived service quality in luxury hotels.

Guest online review is a personal opinion review after booking/experiencing a service/product at a property. His/her review might affect potential guests' expectations and choices for a place to stay. In this research area, the guest review online is associated with guest feedback, management response, and users' decision-making. Traditionally, the hotel web site, online feedback site (i.e., Trip Advisor, Yelp), OTAs site (i.e., Agoda.com, Booking.com, Expedia.com), and social networking site (i.e., Facebook, Twitter) are accumulated as online platforms that customers can look for preferred information and share their comments, reviews and photos on purposes. On the positive sides, they would keep relying on online reviews to decide to book a hotel (Mauri & Minazzi, 2103) (Mauri & Minazzi, 2103); or be happy with the hotel management response to the guest's online comments (Li, Peng et al., 2017a) (Li, Peng et al., 2017a); and vice versa. Moreover, hotel managers take action on each review and observe consumer behaviors in their purchasing process to boost the great technological advance of the online distribution channels.

In the tourism and hospitality service industry, service quality is such a core value to evaluate whether performance meets customer's expectations (Rianthong et al., 2016) (Rianthong et al., 2016), (Zhao et al., 2018) (Zhao et al., 2018). The perceptions of quality dimensions formed the basis of a person's overall quality perception and predicted one's intended behaviors (Boulding et al., 1993) (Boulding et al., 1993). In this regard, it explained a 93% proportion of variance in word-of-mouth communications and 85% of purchase intentions (Alexandris et al., 2002) (Alexandris et al., 2002). For instance, if service quality meets customers satisfaction with the service, they will keep relying on online reviews to decide to reserve a room at a hotel (Mauri & Minazzi, 2013a) (Mauri & Minazzi, 2013a) or be willing to recommend a hotel to others refer to intangible aspects of their hotel stay (Berezina et al., 2016) (Berezina et al., 2016). Consequently, a 1% increase in online customer rating increases sales per room up to 2.68% in Paris and up to 2.62% in London, but higher stars do not increase the sales (Ogut & Tas, 2012) (Ogut & Tas, 2012). Unfortunately, it neglects to explain why former guests' rating scores can influence customer engagement and satisfaction and play

a significant role in hotel management's sales volume and marketing association. Therefore, as the intention is determined by customer satisfaction toward the service quality and it is hypothesized:

$H_1$: The service quality is positively associated with customer satisfaction in luxury hotels.

### *Guest feedback*

Currently, feedback at the base of guest reviews is at the center of any hotel venue. Accordingly, hotel staff needs to read ad listen to the former guests' feedback to minimize it becoming a burden. Some Scholars suggested that guest online feedback is likely reliable (Cheung & Thadani, 2012) (Cheung & Thadani, 2012). The factors of guest feedback affecting customer satisfaction behind Trip Advisors' rating toward customer satisfaction evaluation had significant interaction with visitors' characteristics, including nationality and characteristic destination (Radojevic et al., 2017) (Radojevic et al., 2017).

In addition, the value for money, sleep quality, and overall service were issues that affect each and every one of online reviewers, such as key 'subjective" aspects of their degree of satisfaction on hotel service (Radojevic et al., 2017) (Radojevic et al., 2017). Any positive online reviews engaged the key factors of service quality evaluation and customer satisfaction (Mauri & Minazzi, 2013a) (Mauri & Minazzi, 2013a). Moreover, customers are willing to pay more to have that service (Fuentes, 2016). On the contrary, any negative comments caused by the unpleasant experience can hurt the hotel's reputation and customer's purchasing intention (Wei et al., 2013) (Wei et al., 2013).

The OTA websites perform better in most aspects, while hotel websites are only competitive in website quality (J. N. Liu & Zhang, 2014) (J. N. Liu & Zhang, 2014). If a hotel were listed as one of the best hotels in a well-known online travel community (i.e Trip Advisor) with high rating online reviews, the hotel would be more likely to have higher booking intentions in contrast (L. Casaló a et al., 2015) (L. Casalóa et al., 2015). Consequently, social internet platforms (i.e., Trip Advisor, Facebook, Instagram, and Twitter, etc.) involve peer-to-peer communication channel to one and raises healthy engagement between clients and hoteliers. Social media's minus is considerable when displeasure ideas have expressed leading unhealthy, influencing reputation toward the hotel service quality. Especially customers could revenge on the hotel's reputation if there is an unpleasant experience (Wei et al., 2013) (Wei et al., 2013). Various approaches have been proposed to customer feedback in online hotel reviews. However, there are still some critical issues concerning hoteliers' vital technology readiness to study customer satisfaction and engagement behaviors, enabling them to control the service delivery. Thus, it is posited

$H_2$: The feedback of customer has a positive effect on service quality in the luxury hotels

### Hotel management response

Reading and responding to guest reviews online is a key task for hotel management. The online users are keen on hotel management's responses, especially to those in case of negative reviews. Therefore, hoteliers must apologize to dissatisfied customers, change concepts, and heighten the positive of possible. Accordingly, online platforms' power counts on the hotelier's ability to expand word of mouth (WOM), e-WOM, and social effect among travelers in various tourist destinations worldwide. Besides, management response impacted the readers' likelihood of drawing negative but potentially erroneous inferences (Sparks et al., 2016) (Sparks et al., 2016).

Additionally, the hotel management response is counted on the frequency, speed, and length of content to achieve effective customer interactions (Li, Penga et al., 2017b) (Li, Penga et al., 2017b). However, destination marketers needed to track online chats and real-time reviews on multiple platforms (Soyoung & Busser, 2018) (Soyoung & Busser, 2018). Consequently, a growing body of literature has examined this complicated issue of guest online feedback to enhance e-service quality. However, the previous studies lack the essential for hotel sales and their accessibility to potential customers based on customer satisfaction and engagement behaviors. Therefore, this problem can be posed as the below hypothesis:

$H_3$. The response of hotel management has a positive effect on service quality in luxury hotels.

### Decision-making process

A customer's decision-making process is affected by both internal parts (e.g., customer's personality, attitude, belief, motivation, and intentions); and external components (i.e., time, marketing mix, variety of choices, and others' feedback, etc.) (Sirakaya & Woodside, 2005) (Sirakaya & Woodside, 2005). This concept gives more opportunities for tourists to seek the most suitable alternative possible. Therefore, hotel management needs to study the customer's decision-making process in choosing a hotel to stay.

Recently, the widespread application of Internet technology has changed the way consumers choose and book hotels. This, in turn, led to affect the buying decision process (i.e., customers decide to make a room booking with the hotel because of the guest positive online reviews (Mauri & Minazzi, 2103)

(Mauri & Minazzi, 2103)). Findings from two experiments in Germany and Macau indicated that review valence significantly affects hotel booking intention; afterward, the reader-reviewer demographic similarity moderates the effect (Chan et al., 2017) (Chan et al., 2017). The e-WOM has been used as a low-cost tool to acquire new customers and increase consumers' recommendations (Reimer & Benkenstein, 2018) (Reimer & Benkenstein, 2018). Nevertheless, consumers were likely influenced by early negative information, which led to a negative set of reviews overall (B. A. Sparks & Browning, 2011) (B. A. Sparks & Browning, 2011).

Certainly, negative or positive ideas are now quickly posted by searchers on the Internet regarding their subjective perception of experiences and assessment. The online review has been especially posted by a well-known online travel community (e.g., Trip Advisor) would be a higher booking intention result (L. Casaló a et al., 2015). Thus, as part of the decision-making phase of choice selections, prospective customers could come to the prior purchasers' zone to get information before purchasing. Consequently, more customers are willing to rely on e-WOM as an online key to information for particular products or services. However, for future users, this report fills the gap of an existing theory that customers' decision-making process might enhance or detract customer satisfaction from a hotel brand, and in turn, affect the establishment's reputation. Therefore, the following hypothesis is proposed:

$H_4$. The process of decision-making of customer has a positive effect on service quality in luxury hotels.

### *Customer Engagement Behaviors (CEBs)*

CEBs are defined as customer behaviors that *"go beyond transactions and maybe specifically defined as a customer's behavioral manifestations that have a brand or firm focus, beyond purchase, resulting from motivational drivers"* (Doorn et al., 2010) (Doorn et al., 2010). As one of the early attempts, the present study explored one particular manifestation of CEBs in the hospitality setting: posting online hotel reviews. Accordingly, customers enjoy positive CEBs evaluation more than negative CEBs (Wei et al., 2013) (Wei et al., 2013). They would repurchase services based on previous customers' online feedback, including trust, perceived ease of use, perceived usefulness, and enjoyment (Chiu et al., 2009) (Chiu et al., 2009). For instance, customers would say negative things about the hotel to others if not satisfied with the hotel's service quality; on the contrary, they would make positive online reviews if they were satisfied with the hotel's service quality (Alexandris et al., 2002). In some cases, the satisfaction varied based on the various combinations of recovery measures (i.e., respondents expressed higher satisfaction with the service when

a 50% refund was given) (Sparks & Kennedy, 2001) (Sparks & Kennedy, 2001). Otherwise, the customer would switch to the hotel's competitor (Keaveney, 1995) (Keaveney, 1995).

Customer engagement contributes to higher customer loyalty levels in Indian hotels (Rather & Sharma, 2017) (Rather & Sharma, 2017). Moreover, it directly affects tourists' satisfaction and loyalty in Malaysian hotels (Rasoolimanesh et al., 2019) (Rasoolimanesh et al., 2019). However, the downside of the associations of engaging consumers and technology (Chathoth et al., 2014) (Chathoth et al., 2014) was examined in some contexts (Harrigan et al., 2017), but in the luxury hotel and tourism service. This absence would be fulfilled in this study. Thus, it is posited

$H_5$: The service quality has a positive effect on customer engagement behaviors in luxury hotels.

### *Brand loyalty*

Brand loyalty is important to reduce the production cost and market the product, increasing the profit margins. It occurs when a customer keeps going to purchase a product produced by the same brand instead of a substitute product provided by a competitor (Kotler, 1984) (Kotler, 1984). It has been the main focus of strategic marketing planning and a sustainable competitive advantage through marketing efforts (Dick & Basu, 1994) (Dick & Basu, 1994).

In the hotel and travel context, "*the service brand loyalty can be strengthened not only through the service consumption experience but also through customer engagement beyond the service encounter*" (So et al., 2014). Loyalty was attributed to satisfaction with service quality (Ribbink et al., 2004)(Ribbink et al., 2004). Nevertheless, many providers' loyalty was not a reasonable goal to the nature of the product category or consumer disinterest (Oliver, 1999) (Oliver, 1999).

In addition, different perspectives on loyalty were based on the different frameworks for understanding customer loyalty (Uncles et al., 2003) (Uncles et al., 2003). Therefore, consumer-based brand equity has been found in luxury hotels (Liua et al., 2017) (Liua et al., 2017) to boost loyalty, trust, and brand evaluations (Harrigan et al., 2017). Accordingly, customers are willing to pay more for a hotel booking based on the hotel star rating and former customers' score (Ogut & Tas, 2012) (Ogut & Tas, 2012). Customer engagement was analyzed through data from 298 Taiwanese respondents as the main determinant of brand loyalty in the tourism social media context (M. Li et al., 2020) (M. Li et al., 2020). It is likely to suggest the higher customer engagement in the tourism industry, the bigger ability to maintain sustainable

tourism. Therefore, in this study, the particular brand loyalty might consider as sustainable tourism in the online context.

However, the impact of online reviews on hotel booking intentions was emphasized by the customer's perception (B. A. Sparks & Browning, 2011) (B. A. Sparks & Browning, 2011) and intention to be loyal to the hotel brand (So et al., 2014). Six hundred questionnaires used to study hotel service quality in Beijing in 2016 found that the customer satisfaction degree impacts hotel guest loyalty and revisit intention (Luo & Qu, 2016) (Luo & Qu, 2016). Besides, in an increasingly networked society where customers can interact with others and venues through social networks, CEBs were considered a behavioral manifestation toward the brand beyond transactions (Verhoef et al., 2010) (Verhoef et al., 2010). In addition, the online social media interaction was recognized as a sale and marketing tool to achieve higher levels of customer engagement and loyalty (Vo et al., 2019a) (Vo et al., 2019b). A survey of 170 hotel guests in India in 2020 confirmed the positive influence of customer service experience on satisfaction and loyalty (Paulose & Shakeel, 2021) (Paulose & Shakeel, 2021). There is still considerable ambiguity with regard to the combined factors of guest reviews online influencing customer satisfaction toward service quality in which have a knock-on-effect on CEBs and brand loyalty. Therefore, this study proposes the hypotheses as below:

$H_6$: Customer satisfaction is positively associated with CEBs in luxury hotels.

$H_7$: CEBs are positively associated with brand loyalty/sustainable tourism with luxury hotels.

Till now, reason-based arguments and scientific reports have not been shown and published much regarding the impacts of guest online reviews in the Vietnamese hotel and tourism context. It is; therefore, less likely industry' facts and figures in the measurement of customer satisfaction for tourists perceive hotel service quality, afterward lead to repurchase service in the future, but reviews and comments via hotel' website, OTA sites, and social channels.

## RESEARCH METHODOLOGY

The study's objectives were to investigate the impact of guest online reviews on service quality that the hotel enterprises might face to achieve customer satisfaction and identify according to customer satisfaction that supports the degree of customer engagement behaviors and the business can render to help them gain brand loyalty. The current research proposes and tests a conceptual model of the guest online review influencing customer satisfaction toward service quality in Vietnam's luxury hotels. The proposed model was drawn and

developed from previous literature highlighting the importance of guest reviews on the internet before deciding "where to stay," i.e., Guest online reviews include guest feedback (Cheung & Thadani, 2012) (Cheung & Thadani, 2012), (Wei et al., 2013) (Wei et al., 2013), (Fuentes, 2016); management response (Mauri & Minazzi, 2103) (Mauri & Minazzi, 2103) (L. Casalóa et al., 2015) (Chan et al., 2017); decision-making process of the customer (L. Casalóa et al., 2015) (Chan et al., 2017), etc. (details as Table 1). The proposed model includes six constructs: guest online reviews (03 independent variables); and 03 independent variables: service quality, customer satisfaction, CEBs, and brand loyalty. The proposed conceptual framework of customer satisfaction toward guest online review is illustrated in Figure 1.

This study associated with quantitative research. Accordingly, the knowledge is based on objectives, and the real problem exists. Therefore, the reality is generalized by theory testing and verified hypotheses without the author's beliefs and biases (Creswell, 2009) (Creswell, 2009). Survey research was chosen because it is one of the most practical hotels, tourism, and applied social research to collect information about how things are at a defined-phenomenon. Accordingly, a survey is well suited to this descriptive research in the cross-sectional data. It is used to examine a situation by determining important dimensions associated with respondents' satisfaction, attitudes, experiences, etc.

The quantitative data were collected by a questionnaire that included a Trip Advisor sample using to illustrate the importance of both positive and negative guest reviews on the internet during the pre-travel step to decide "where to stay". The data collection technique was in person and online approach via the researchers' Facebook and e-mail accounts. The open questions in this questionnaire help generate the ideas of the respondents based on guest reviews online.

This study's sample framework is on the luxury hotels (from 4-star to 5-star hotels) in Vietnam. The International and domestic tourists to VN have risen 15.5 mils and 80 mils, respectively, with total revenue of VND700 trillion, approximately $US32 billion in 2019 (Anonymous, 2019a). The volume was based on the markets listed from the most to the least International visitors to Vietnam as Asia, Europe, America, Oceania, and Africa (Anonymous, 2019b)(Anonymous, 2019b). The mainland Chinese was the majority of Asians in 2018 with air transportation (23.9%) (PATA, 2019) (PATA, 2019).

There are 391 luxury hotels/resorts (4-star and 5-star) in Vietnam, including 261 4-star hotels with 33,764 rooms, 118 5-star hotels with 34,444 rooms, and 12 high-end resort apartments (VNAT, 2018) (VNAT, 2018). The luxury hotels and resorts grew at 19% in total room revenue (CCIFV, 2018) (CCIFV, 2018). Therefore, the hotel industry is growing healthily by International hotel groups, State companies, and local Joint Stock companies.

**Table 1.** Key references of the constructs and measurements.

| Code | Statements | Key references |
|---|---|---|
| | **Guest Online Feedback** | |
| OF1 | The guest online feedback is reliable | (Cheung & Thadani, 2012)(Cheung & Thadani, 2012) |
| OF2 | Customers can revenge the hotel's reputation if there are unpleasant experiences | (Wei et al., 2013)(Wei et al., 2013) |
| OF3 | The negative comments can be an outlet for customers to hurt the hotel's reputation after an unpleasant experience | |
| OF4 | The price I paid was too high due to booking at a hotel based on hotel star rating and scores awarded by former customers | (Fuentes, 2016) |
| | **Hotel Management Response** | Key references |
| MR1 | The hoteliers frequently respond to guest online comments | (Li, Peng et al., 2017a)(Li, Peng et al., 2017a) |
| MR2 | The hoteliers promptly respond to guest online comments | |
| MR3 | The hoteliers' response is clear and effective communication to engage customers | |
| | **Decision-Making Process** | Key references |
| DM1 | I decide to make room booking with the hotel because of the guest positive online reviews | (Mauri & Minazzi, 2103)(Mauri & Minazzi, 2103) |
| DM2 | The online rating lists are more useful and credible when published by well-known online travel communities (e.g., TripAdvisor). | (L. Casaló a et al., 2015) |
| DM3 | The higher booking intentions result if the list is published on a well-known online travel community (TripAdvisor) | |
| DM4 | Online review valence significantly affects hotel booking intention | (Chan et al., 2017) |
| | **Overall Service Quality** | Key references |
| SQ1 | Service quality evaluation remains key factors that stimulate positive online customer reviews | (Mauri & Minazzi, 2103)(Mauri & Minazzi, 2103) |
| SQ2 | Customers can use the hotel stars as a tool to choose hotel establishment and the price they are willing to pay | (Ogut & Tas, 2012)(Ogut & Tas, 2012) |
| SQ3 | Individual reviewers play an important role in more "subjective" aspects of service, such as value, quality of sleep, and service | (Radojevic et al., 2017)(Radojevic et al., 2017) |
| | **Overall Customer Satisfaction** | Key references |
| CS1 | I am satisfied with the guest's on line reviews | (Cheung & Thadani, 2012)(Cheung & Thadani, 2012) |
| CS2 | I keep relying on online reviews to decide booking a hotel | (Mauri & Minazzi, 2103)(Mauri & Minazzi, 2103) |
| CS3 | I am happy with the hotel management response to the guest's online comments | (Li, Peng et al., 2017a)(Li, Peng et al., 2017a) |
| | **Customer Engagement Behaviors** | Key references |
| CEB1 | I would say negative things about the hotel to others if I am not satisfied with service quality at the hotel | (Alexandris et al., 2002) |
| CEB2 | I would make positive online reviews if I am satisfied with service quality at the hotel | |
| CEB3 | I would ask for refund if I am not satisfied with service quality at the hotel | (Sparks & Kennedy, 2001)(Sparks & Kennedy, 2001) |
| CEB4 | I would repurchase service based on online feedback provided by previous customers | (Chiu et al., 2009)(Chiu et al., 2009) |
| CEB5 | I would switch to hotel's competitor if I experienced unfair prices | (Keaveney, 1995)(Keaveney, 1995), (Voss et al., 1998)(Voss et al., 1998) |
| | **Hotel Brand Loyalty/Sustainable Tourism** | Key references |
| BL1 | I'm willing to pay more for a booking at a hotel based on hotel star rating awarded by former customers | (Ogut & Tas, 2012)(Ogut & Tas, 2012) |
| BL2 | I'm willing to pay more for a booking at a hotel based on scores awarded by former customers | (Ogut & Tas, 2012)(Ogut & Tas, 2012) |
| BL3 | I would make the bookings with its price policy in the future | (Fuentes, 2016)(Fuentes, 2016) |

The term "luxury hotels" has been applied to this report. It is used for hotels and resort zones meeting the requirements and policy of the VNAT board toward the star rating of service standard focused on the 4- and 5-star hotels and resorts. Luxury hotels offer customers the outstanding quality of hotel

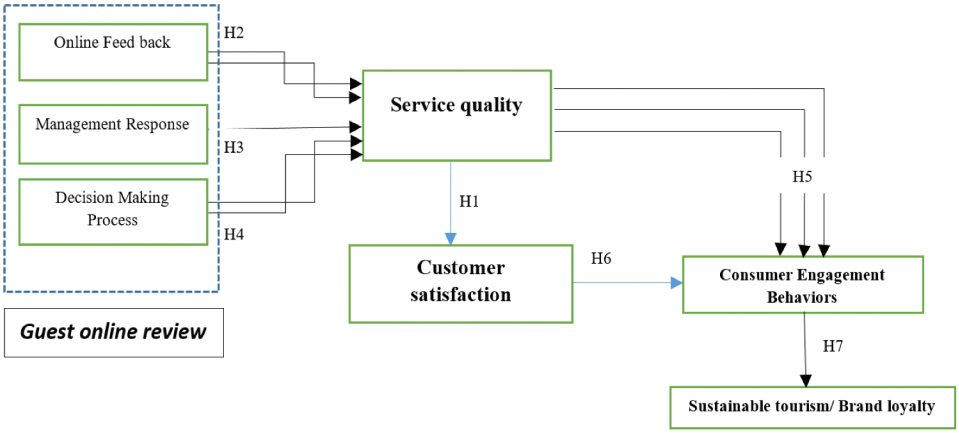



**Figure 1.** Proposed conceptual framework of customer satisfaction toward guest online review.

service to satisfy and delight hotel guests, including personalized service, spa, fine dining services, butler services, and other fancy services.

The particular customers of those luxury hotels are high-paying guests, and in return, they expect to have high standards such as excellence in service, in rooms, in dining, and in everything else a good hotel offers (Trip savvy, 2019) (Trip savvy, 2019) despite the purposes of stay for business or leisure. They want to be the very important persons (VIPs) and, consequently, are straightforward to ask for higher requirements and/or ask for hotel' compensations for any troubles or complaints related to the quality of service or facility provided by hotels. Therefore, the hotels need to stay focused to deal with guests, especially with guests' complaints, such as read guest feedback, keep an eye on the website of OTAs and hotel venues, respond to comments, etc.

Some researchers have studied the influence of online ratings and reviews on hotel booking consideration using mixed methods (Gavilan et al., 2018) (Gavilan et al., 2018); (Ahmad & Sun, 2018) (Ahmad & Sun, 2018). A field survey was delivered to achieve the stated research objectives. A questionnaire was developed with three sections. Following previous studies, the maximum is a 12-month period for respondents who had stayed a room from 4-star to 5-star hotels within Vietnam to recall their experiences (Law & Hsu, 2006) (Law & Hsu, 2006) to imply the role of understanding the exiting and potential customers' needs (Law & Leung, 2000) (Law & Leung, 2000).

The open questions were formulated in the first section to identify eligible respondents and customer cognition of guest online reviews toward luxury hotels. The answers enabled respondents to proceed with the survey. The second section probed the seven Likert-scale included in the proposed model (e.g., guest online feedback, management response, decision-making process, hotel website quality, customer satisfaction, CEBs, and brand loyalty). Table 2 shows the constructs with multiple items that followed previous

**Table 2.** Our customers' demographic and experiences of online hotel booking and searching information.

| Descriptions | N = 384 | % |
|---|---|---|
| **Gender** | | |
| Male | 119 | 31 |
| Female | 265 | 69 |
| **Daily internet usage** | | |
| <2 hours | 5 | 1.3 |
| 2–4 hours | 124 | 32.3 |
| >4 hours | 255 | 66.4 |
| **Booked the rooms via** | | |
| Online Travel Agent | 161 | 42 |
| Hotel website | 83 | 22 |
| Travel operator | 35 | 9 |
| Other (direct contact to hotel included phone, walk-in, sale persons) | 105 | 27 |
| **Booked the rooms via websites of OTAs** | | |
| Never | 163 | 42 |
| 1–3 times | 176 | 46 |
| >3 times | 45 | 12 |
| **Age bracket** | | |
| 16–25 | 74 | 19 |
| 25–35 | 152 | 40 |
| 36–45 | 122 | 32 |
| >45 | 36 | 9% |
| **Personal monthly income** | | |
| <$500 | 96 | 24.9 |
| $500-$1000 | 156 | 40.4 |
| $1000-$1500 | 61 | 15.8 |
| $1500-$2000 | 35 | 9.1 |
| >$2000 | 36 | 9.8 |
| **Education** | | |
| Vocational College | 28 | 7 |
| Bachelor Degree | 138 | 36 |
| Post Graduate | 218 | 57 |
| **Current occupation** | | |
| Staff of hotel/tourism establishment | 100 | 26.0 |
| Freelancer | 17 | 4.4 |
| Lecturer in hotel and tourism management | 115 | 29.9 |
| Office staff | 69 | 18.0 |
| Businessman | 17 | 4.4 |
| Other | 68 | 17.2 |
| **Working experience** | | |
| <1 year | 36 | 9.8 |
| 1–3 years | 75 | 19.4 |
| 3–5 years | 35 | 9.1 |
| 5–10 years | 111 | 28.8 |
| >10 years | 127 | 32.9 |

studies closely. The questionnaire in English originally consists of a seven-point Likert scale representing the measurement of all variables reflected in the conceptualization will serve as a tool to collect data *(1. Strongly Disagree, 2. Moderately Disagree, 3. Disagree, 4. Neutral, 5. Agree, 6. Moderately Agree, 7. Strongly Agree)* and to expand the number of choice-points which would increase sensitivity without damaging scale reliability (Cummins & Gullone, 2000) (Cummins & Gullone, 2000). The English questionnaire forms were delivered to International travelers. The questionnaire was translated into Vietnamese for Vietnamese respondents and been reviewed by English

expertise. The data has been translated into English (back-to-back translation) for coding and analysis. The third section measures demographic characteristics, including gender, sex, age group, time of internet usage, education level, continents, monthly personal income, working experiences, etc.

The survey employed 400 hotel and tourism professionals who have been experiencing, training, and knowing about the service quality, facility, and customer-related benefits in the hotel and tourism industry. Especially, they have recently been living in Ho Chi Minh City. This area is the most majority of hotels in 91 luxury hotels within Viet Nam (Anonymous, 2017b) (Anonymous, 2017b). Moreover, it is well known because of its size and most developed in service, economy, and education industry, especially in hotels and tourism vocational schools and universities. Furthermore, the respondents are from the Bachelor's degree level, early adaptors, brand influencers, and social media drivers. The researchers invited from 05 to 10 volunteers to get into a room and delivered the survey. We explained the meaning of the study through a structured 03-open question and 16 scaling-question surveys. The Vietnamese survey version was delivered to take feedbacks from Vietnamese respondents then translated into English by English expertise for accurate coding and analyzing data strictly. The response time is from 7 to 10 minutes for each respondent providing the answers from 2017 Sep 15 to Dec 29. The 16 samples were filtered out because the participants did not complete the questionnaire or rated all measurements at the same scores. Thus, there are finally 384 valid samples.

According to Hair's sample size (2006), if the population is above one million sample units and the population is unknown, the appropriate sample size is 384 respondents with the standard of 95% confidence level and a 5% margin of error. The IBM SPSS 22 tool was used to analyze participants' demographic characteristics and engagement behaviors on service quality when making online/offline reviews and their experiences on hotel booking channels. Each factor was calculated the Cronbach's Alpha (CA) coefficient value (>0.7) to measure the internal consistency, i.e., reliability of the measuring instrument of the questionnaire. The questionnaire's validity was followed by Kaiser Meyer Olkin (KMO) Bartlett's test of Sphericity which can vary from 0 to 1, indicates the degree to which each variable in a set is predicted without error by the other variables. Hair et al. (2006) suggested to accept a value above 0.5 or between 0.5 and 0.7 is mediocre, and values between 0.7 and 0.8 are good. The relationship between the dependent variable and a set of multiple independent variables, linear regression analysis was conducted. The estimation of regression coefficients, standard error of the estimates, t-tests, and significance trained the coefficient (Priya & Sharma, 2015) (Priya & Sharma, 2015). ANOVA test was conducted to determine the significant differences between the means of three or more independent variables to undertake different tasks and measure the outcome of the dependent variables (Priya &

Shruti, 2015) (Priya & Shruti, 2015). The Model Summary included multiple correlation coefficient R and its Square ($R^2$); and the adjusted version of this coefficient as summary measures of the model fit (Priya & Sharma, 2015) (Priya & Sharma, 2015). The application of simple linear regression (var: 1 dep., 1 indep.) has formula: $y_i = 1a + X_i b + \varepsilon_i$ to multiple linear regression (var: 1 dep., >2indep.): $y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \hat{\beta}_2 X_{i2} + \varepsilon_i$ by performing a regression analysis on this survey data helps us determine whether these variables have impacted overall attendee satisfaction, and if so, to what extent. The equation of the regression line was: $\hat{y}_{zi} = \hat{\beta}_{z1zi1} + \hat{\beta}_{z2zi2} + \ldots . + \hat{\beta}_{zkzik}$. It informed us about which elements of the sessions are well received and where we need to focus, so the attendees are more satisfied in the future.

## EMPIRICAL FINDINGS & DISCUSSIONS

We recruited 265 women (69%) and 119 men (31%). Their ages are not significantly different in those who have experienced and worked in the four and five-star hotels in Vietnam. The respondents used to surf the Internet below <2 h (1%), 2–4 h (32%), >4 h (67%). This data analyzed whether the observed relationships between respondents' internet usage daily and online booking behaviors on the websites were attributable to genetic or environmental factors. Some incomplete answers were eliminated, resulting in 384/400 completed questionnaires (96%). Qualitative data with 03 open questions explored the CEBs. The respondents' characteristics, which may influence their engagement intentions on service quality for sustainable development, influence their satisfaction level. The expected indicator as "Guest online review" was illustrated in a case study of the Trip Advisor with two different guest comments online. The respondents will firstly be asked to read the illustrated comments, then stated their degree of agreement with the statements provided toward their perception of guest' reviews.

### *Qualitative data on the survey*

The driving factors of guest perception influencing the customer satisfaction levels through perceived service quality established by open questions: "*You experienced the service, what would you do in response to the levels of your satisfaction in relation to hotel service quality? List in order of the most important to lower (from one to four)*". Adapted from Braun and Clarke (2006), the qualitative data has been analyzed by some phases, including familiarizing the authors with data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and eventually producing the result with five dimensions (Braun & Clarke, 2006) (Braun & Clarke, 2006). Consequently, the results showed that (1) the 25% of "self-protection" (Radojevic et al., 2017) (Radojevic et al., 2017) dimension shows

guests' reaction on "complain to hotelier" (Alexandris et al., 2002) and "cancel and switch to other hotels" (1%) (Keaveney, 1995) (Keaveney, 1995); (2) the 23% of "revenge" (Wei et al., 2013) (Wei et al., 2013) dimension has shown guest disappointment to express on "online review" and "offline" – WOM at 17%; (3) the 17% of "repurchase" (Chiu et al., 2009) (Chiu et al., 2009) dimension has shown the positive guest reaction on the satisfaction mood on "recommend to others"; (4) of 'return" (Fuentes, 2016) dimension to use the service of the hotels at 3% and (5) last one is rewarded/other at 15%.

To conclude, the qualitative data collected the various ways and levels of customer reaction while receiving services at the hotels. Firstly, they would mostly protect themselves against dissatisfaction by complaining, canceling the stay, or even switching to other hotels. Secondly, they could take revenge for their defeat of the service through WOM or e-WOM. Last but not least, they would repurchase the service in advance or recommend the hotels to others. In another action, they could give tips and keep in touch with the hotel venue for further association.

### *Quantitative data of the survey*

The demographic profile of the participants is shown in Table 2. The majority of 384 respondents were female (69%), young people with higher daily internet usage than 4 hours (66.4%). In addition, the participants' online hotel booking experienced, mostly in OTAs channel (42%), the rate of booking over the website of OTA in total was 58% in which via hotel websites at 22%, travel operator at 9%, and direct contact to hotels such as via the phone, walk-in, and sale in person at 27%. The respondents mostly booked the room via OTA websites from one to three times at 46%, more than three times were at 12%, but people who prefer other booking channels at 42%. Especially, the applicants are experienced and trained in the hotel and tourism industry at 55.7% from positions of Staff, Supervisor, Manager and General Manager (44%, 14.8%, 19.9%, and 1.8%, respectively). This batch of respondents is the strict and high expectation from the value of service quality. The respondents mostly worked from above 5 years experience at 61.7%. The personal income monthly from above US$500 was 75.1% enable respondents to use service from luxury hotels.

Vietnam Investment Review reported that 4.1 million Chinese tourists were taken over the total of 7.9 million International tourists to Vietnam in the half-year of 2018 (VNAT, 2018). To expand this batch of travel to VN, the Government has implemented the e-visa system and exempted visa requirements for European tourists such as France, Germany, United Kingdom, Italy, and Spain who are staying for less than 15 days. They also have extended the days-off, promoted festivals and events to Domestic travelers. So, despite the purposes of leisure or business trips, the numbers of both local people

(domestic tourists) and foreigners (inbound tourists) are gradually increasing. To a certain extent, an increase in Vietnamese income makes a greater chance for domestic travelers' demand in using service of luxury hotels.

***(Source: The results of descriptive analysis from the SPSS tool)***

The validity of this research was firstly ensured by the construction of the questionnaire with 59 participants of the Pilot test and then tested KMO and Bartlett's Test of Sphericity to ensure that factor analysis is appropriate. The overall Mean score items were from 4.76 for MR2 *(Hotel promptly responses guest online comments*) to 6.15 for SQ11 (*Service quality evaluation stimulates positive online review*). This analysis indicated that participants somewhat use these criteria when selecting hotel booking channels toward management response (Li, Peng et al., 2017a) (Li, Peng et al., 2017a) and positive online reviews (Cheung & Thadani, 2012) (Cheung & Thadani, 2012). The CA for each attribute showed that the OF1 independent variable *(Online feedback is reliable)* had a coefficient of correlation of 0.239 < 0.3. CA value was significant if DK1 Item Deleted was 0.854 > 0.768. Therefore, the authors decided to exclude the OF1 variable to increase the reliability of the scale. Rerunning the test, we had the results of 24 out of 25 attributes >0.7 (from 0.733 to 0.904). The KMO of independent variables were 0.678, and dependent variables were 0.73 with both Sig Barlett's Test = 0.000 ($p < .05$). Consequently, the questionnaire was reliable to run for further analysis with 384 valid samples.

We found that the results of the CA and Exploratory factors (KMO) of each attribute by category were analyzed. The CA values for all of the factors were acceptable (95% confidence interval) (Hair et al., 2006). The overall CA of variables was from 0.729 (customer satisfaction) to 0.899 (Management response). The overall KMO of all dependent variables (i.e., service quality, CEBs, brand loyalty, and customer satisfaction) was 0.693. The KMO ratio of all independent variables (i.e., online feedback, management response, and decision-making) was at 0.666, with Sig of Chi-Square for both factors was 0.000 ($p < .005$). The rotated component matrix results showed that 16 observed variables had a loading factor greater than 0.5. The tests proved that all factors analysis were appropriate to be conducted (see Table 3).

***(Source: The results of SPSS tool)***

The Pearson Correlation coefficient was calculated to assess whether the relationships (weak/strong) among median variables. The Pearson Sig. (2-tailed) correlation among OF and DM independent variables with SQ, CEB, BL and CS dependent variables were less than 0.05. Thus, there were linear relationships among these independent variables with SQ, CEB, BL, and CS dependent variables. The correlation between DM and SQ was strongest with r = 0.584, and between MR and BL, the correlation was the weakest (r = 0.008). Therefore, hotel management must notice customer's decision-

**Table 3.** The indicator reliability of the items and constructs.

| Guest online review | | | Mean | Cronbach's Alpha | KMO | Notes |
|---|---|---|---|---|---|---|
| Guest online feedback (OF) | OF1 | Online feedback is reliable | 5.391 | 0.702 | | Exclude OF1 after rotated component matrix |
| | OF2 | Negative comment can revenge hotel' reputation | | | | |
| | OF3 | Negative comment can hurt hotel' reputation | | | | |
| | OF4 | Paid higher price based on online review | | | | |
| Management response (MR) | MR1 | Hotel frequently responses guest online comments | 4.948 | 0.901 | | Independent variables |
| | MR2 | Hotel promptly responses guest online comments | | | | |
| | MR3 | Hotel' response is clear to engage customer | | | | |
| Decision making (DM) | DM1 | Decide to book room because of positive online review | 5.992 | 0.853 | | |
| | DM2 | Online rating lists are more useful and credible on Trip Advisor | | | | |
| | DM3 | Higher booking intentions result on Trip Advisor | | | | |
| | DM4 | Online review affects hotel booking intention | | | | |
| Service quality (SQ) | SQ11 | Service quality evaluation stimulates positive online review | 6.023 | 0.736 | 0.691 | Dependent variables |
| | SQ12 | Customer is willing to pay based on hotel star rating | | | | |
| | SQ13 | Individual reviewers are subjective on value, sleep quality, and service | | | | |
| CEBs | CEB11 | Say negative if not satisfied with service quality | 5.185 | 0.733 | | |
| | CEB12 | Make positive online reviews if satisfied with service quality | | | | |
| | CEB13 | Ask for refund if not satisfied with service quality | | | | |
| | CEB14 | Repurchase service based on online feedback | | | | |
| | CEB15 | Switch to other hotel if experienced unfair price | | | | |
| Brand loyalty (BL) | BL11 | Be willing to pay more to book a room based on hotel star rating | 5.092 | 0.738 | | |
| | BL12 | Be willing to pay more to book a room based on score rating | | | | |
| | BL13 | Make booking with hotel' price policy in the future | | | | |
| Customer satisfaction (CS) | CS11 | Be satisfied with guest online review | 5.162 | 0.733 | | |
| | CS12 | Keep relying on online review to decide booking | | | | |
| | CS13 | Be happy with hotel management' responses on online comment | | | | |

king process enhances e-WOM such a low-cost approach to attract potential customers (Reimer & Benkenstein, 2018) (Reimer & Benkenstein, 2018). However, hotel management needs to keep in mind that negative information on the internet sites significantly impacts sustomers' decision to stay overnight (B. A. Sparks & Browning, 2011) (B. A. Sparks & Browning, 2011). Another point of view, the Pearson correlation between BL and MR was greater than 0.05 at 0.876, so there was no linear correlation between these variables. The MR variable was eliminated when performing linear multiple regression

analysis. Therefore, the independent variable pairs had relatively weak correlations, so there would be no multi-linearity. Eventually, after the Pearson analysis, there were only two independent variables: OF and DM.

Then, we found that the ANOVA analysis results would enable the researchers to determine the difference between variables. The adjusted R2 value of 0.336 showed that the independent variables (OF, DM) introduced into the regression run affected 33.6% of the dependent variable (SQ) variance, while the remaining 66.4% was due to exogenous variables random errors. According to Field (2009), if the Durbin coefficient – Watson (DW) is less than 1 and greater than 3, we need to notice this because of the high probability of the first-order autocorrelation. The value of DW is in the range of 1.5–2.5; there will be no autocorrelation. This is also the standard value we are using today. The output showed that DW = 2.245, which was in the range of 1.5 to 2.5, so there was no autocorrelation occurs. The result of ANOVA reflected statistically significant *p*-value i.e. $p = .000$ ($p < .05$) and F = 98.914. Thus, multiple linear regression models were suitable for data sets and can be used. Sig. Testing of the DM's regression coefficient was less than 0.05, but OF beta was less than 0.05 at −0.236 and Sig. >0.05 at 0.597, so the DM explained the dependent variable, OF was excluded from the model. Therefore, the VIF coefficients of the independent variables were less than 2 at 1.065 so there was no multi-linearity. The regression coefficient DM was greater than 0, and was incorporated into the regression analysis to affect SQ's dependent variable. The histogram of dependent variable: SQ Service quality had Mean = −3.21E-15 was near zero, the standard deviation was 0.997 and closed to 1 to distribute the approximation. Therefore, it can be concluded that: the normal distribution hypothesis of the remainder was not violated. The normal *P*-P plot of regression standardized residual of SQ showed that the distributing points in the distribution of the remainder were centered into a diagonal, thus, assuming the normal distribution of the residual was not violated. SQ's scatter plot showed that the normalization residue concentrates around the ordinate line 0, assuming linear relations were not violated.

Continuously, we analyzed the ANOVA analysis of independent variables (OF, DM) and the dependent variable (CEB). The adjusted $R^2$ value of 0.076 shows that the independent variables (OF, DM) introduced into the regression run affected 7.6% of the variance of the dependent variable (CEB), while the remaining 66.4% was due to exogenous variables and random errors. The Durbin – Watson coefficient was at 1.882, which was not occurring autocorrelation. We found that, the ANOVA reflected statistically significant *p*-value i.e. $p = .000$ ($p < .05$) and F = 16.761. Thus, multiple linear regression models were suitable for data sets and can be used. Sig. Testing the regression coefficient of DM and OF was less than 0.05, at 0.157 and 0.201, respectively. So DM and OF were the explanation of the dependent variables. Both VIF coefficients of the independent variables were no multi-linearity at 1.065. The

regression coefficient of DM and OF was greater than 0 and was incorporated into the regression analysis to affect the dependent variable CEB. Eventually, the histogram of dependent variable: CEB_Consumer Engagement Behaviors had Mean = −5.85E-15 was near zero, the standard deviation was at 0.997 and distributed the approximate approximation.

Therefore, it could be concluded that the normal distribution hypothesis of the remainder was not violated. The normal *P*-P plot of regression standardized residual of CEB showed that the remainder's distribution points are centered into a diagonal, thus assuming the residual's normal distribution was not violated. The scatter plot of CEB showed the normalization residue concentrates around the ordinate line 0, so assuming linear relations were not violated.

### *Hypothesis testing*

Data supported all seven hypotheses. Thus, seven hypotheses, H1 to H7 we had originally been developed in the research hypothesis. Those accepted hypotheses: H1, H2, H3, H4, H5, H6, and H7 correspond to variables of guest online review: Online feedback (OF), Management Response (MR), Decision Making (DM), Service Quality (SQ), Customer Satisfaction (CS), Consumer Engagement Behaviors (CEBs), and Brand Loyalty (BL). In other words, all attributes have made sense in the regression model with a significant *p*-value was 0.000.

The linear regression attempts to model the relationships among variables by fitting a linear equation to observed data. The results confirm the relationships between the variable of interest significantly. The standardized coefficients $\beta$can be obtained from non-standardized using the formula: $\hat{y} = bX + a$, where X is the explanatory variable (OF, MR, and DM) and Y is the dependent variable (SQ).

The standardized coefficients $\beta$can be obtained from non-standardized to the equation of the regression line becomes $\hat{y}_{zi} = \hat{\beta}_{z1zi1} + \hat{\beta}_{z2zi2} + \ldots + \hat{\beta}_{zkzik}$. So the output has its standardized regression equation:

SQ = 0.1236*OF + 0.108*MR + 0.584*DM and CS = 0.468*SQ

CEB = 0.138* OF + 0.173*MR + 0.203*DM + 0.265*CS and BL = 0.276*CEB

Customer satisfaction = 0.468*(0.1236*OF + 0.108*MR + 0.584*DM)

Brand loyalty = 0.276*(0.138* OF + 0.173*MR + 0.203*DM + 0.265*(0.468*(0.1236*OF + 0.108*MR + 0.584*DM))

The findings are highlighted and discussed with respect to the two research questions as follows. Regarding the research objectives, the study confirms that guest feedback online, the response of management, and the customer's decision-making on service quality are important enablers for strengthening the hotel performance on bookers' review on the Internet. Therefore, considering activities such as encouraging bookers' intention on posting the

positive feedback online, guessing and managing decision-making properly for online and offline feedback of users, and avoiding inappropriate response of management to boost satisfaction and loyalty levels of hotel users to hotel brand name.

Figure 2 shows our research model and the results of hypothesis testing. The output indicates that the average level of customer satisfaction toward guest online reviews is positively comprised of perceiving the service quality, including information provided by guest online feedback, the length, speed of management response, and customer cognition of making the online decision process platform. In turn, the low degree of sustainable tourism or brand loyalty regarding the guest online review is the value-added of the volume of customer satisfaction and customer engagement behaviors. However, compared with the Management Response dimension (at the lowest level of regression), the Decision-Making dimension (at the highest level of regression) was 3.7 times as likely to be alert (9%/34%). For future users, the dimension might enhance or detract from a hotel brand, affecting the hotel's reputation (B. A. Sparks & Browning, 2011)(B. A. Sparks & Browning, 2011). Whereas, potential customers may involve in the past purchasers' ideas before buying products/services, such as a part of the decision-making phases of choice selections response (Li, Peng et al., 2017a) (Li, Peng et al., 2017a). Consequently, more and more customers get involved in e-WOM as an online key of information for particular products or services (Cheung & Thadani, 2012) (Cheung & Thadani, 2012).

### *Discussions of the key results*

Our experiments confirm that the guest reviews online are positively associated with customer satisfaction on the perceived service quality. This result is significant only in luxury hotels. Our experiments are consistent with previous

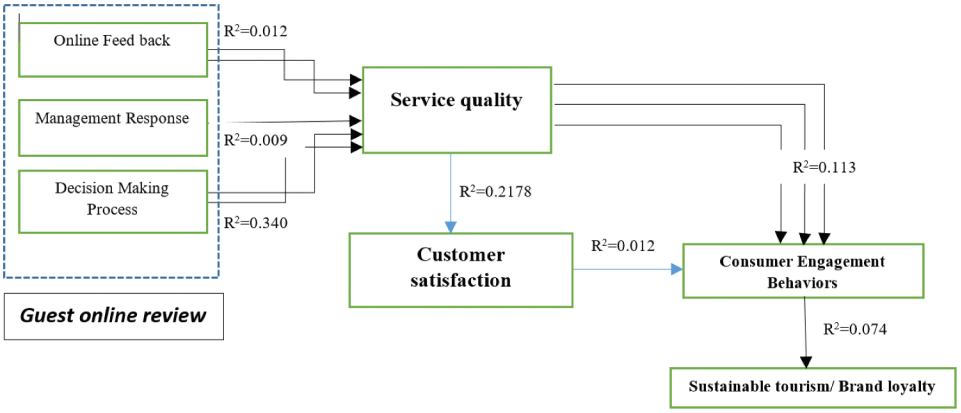



**Figure 2.** Confirmation of our research model.

results. A study by Mauri and Minazzi (2013a) stated that customers' feedback is such a key factor of service quality evaluation and customer satisfaction. The study of Li, Peng et al. (2017a) mentioned that hotel management's response affects the service quality based on customer satisfaction in the online platform. Chan et al. (2017) confirmed the process of decision-making affects personal satisfaction on service quality. However, we found much higher value for guest reviews online with respect to those reported by a study of Soyoung & Busser (2018) to just focus on chats and real-time reviews on multiple online platforms; and by the study e-WOM of Reimer & Benkenstein (2018) to use low-cost tools to increase consumers' intentions and recommendation.

We found that the most striking result to emerge from the data is the decision-making process of customers. It takes the most significant role in generating customer perception, attitude, intention, motivations, and others' feedback. This, in turn, affects the customer satisfaction levels, engagement behavior and develops brand loyalty for the hotel venue.

As hypothesized, our experiments show that guest online review factors (customer feedback, management response, and decision-making process) positively affect CEBs through perceived service quality on luxury hotels. These results differ from some published studies of B. A. Sparks and Browning (2011) about customer perception, and the study of So et al. (2014) focuses on the customer's intention to be loyal to the hotel brand. However, they are consistent with those studies of Rather and Sharma (2017) and Rasoolimanesh et al. (2019) to conclude the interaction of customer satisfaction, CEBs, but not brand loyalty.

## CONCLUSIONS AND IMPLICATIONS

### *Conclusions*

Guest online reviews have significantly become a crucial part of consumer activities on the Internet to study over the world. It is more important for creating and reinforcing consumer engagement behaviors from the supporting of customer satisfaction through the perceived quality of service/product in different dimensions and economic contexts. As the background mentioned above, the present study aimed at identifying factors that improve online communication toward guest online reviews. This afterward increases customer satisfaction, engagement behaviors, and brand loyalty based on luxury hotels' service quality in Vietnam. This is the first study exploring the contributors of service on the online environment and to increase the number of hotel booking guests based on guest reviews.

This research aims to study the contribution of guest online reviews toward the luxury hotels and tourism sector in Vietnam to improve customer

satisfaction and brand loyalty on the online platform. Consequently, the outputs have fulfilled the proposed research questions as below

- Identify the effects of guest online reviews (including guest feedback, management response, and customer decision-making) on customer satisfaction toward service quality in hospitality and tourism enterprises
- To measure customer satisfaction levels affect the customer engagement behaviors
- To confirm the effect of customer engagement behavior on the brand loyalty

### *Implications*

The findings of this study provide important implications for hotel and tourism business organizations. It utilizes the online community as a platform for its consumers to share experiences and give a chance to recover problems, if any, for service suppliers. With this knowledge, the hotel Owners and Sales & Marketing Managers can leverage the guest online review to developing hotel service. Firstly, customers' feedback is significant advice to indicate customer satisfaction and engagement behaviors toward the hotel's perceived service quality. Next, Management response can enhance customer engagement behaviors, especially when there are service-related complaints on the Internet. Also, decision-making is such customer' buying behavior that is significantly affected by previous positive ratings from other users. The "self-protection" dimension is a guest's major reaction to complain and cancel or switch to other hotels toward the CEBs study. Furthermore, the preferred purchasing channel from young online samplings is OTAs sites which can improve 4-star and 5-star hotels with brand images but commission fee for OTAs sites of marketing and maintaining cost.

Additionally, the study supports the absence of existing theories such as guest online review as a generalized consideration for service quality in the hotel setting. The point of view of respondents who are major and experienced in service and operation of Vietnamese luxury hotel context missed out. Therefore, this study offers the theoretical contributions to the existing literature, they are (1) identifying the value of guest reviews online, (2) revealing the most important items of guest reviews online to access customer satisfaction, engagement behaviors, and brand loyalty for sustainable development, and (3) introducing a more comprehensive model for understanding the path of developing customer satisfaction, CEBs and brand loyalty toward guest online reviews on luxury hotels

The influence of online ratings and reviews on hotel booking consideration is inclined with the findings of Gavilan et al. (2018) and N. Vo et al. (2019). In addition, the service quality builds customers satisfaction and, in turn, helps them to keep searching online reviews before booking a room at a hotel (Mauri

& Minazzi, 2013a) (Mauri & Minazzi, 2013a); then, they likely recommend that hotel to potential customers (Berezina et al., 2016) (Berezina et al., 2016). However, the study was focused on the pre-purchase stage, which is the opposite of Ahmad and Sun (2018) results indicated the different points of view in the effects of customer perception on e-WOM and purchase intention toward CEBs in the post-purchase stage. However, our study's output clarified and highlighted the outcome of brand loyalty, which is inclined with studies of Sparks and Browning (2011) and So et al. (2014). Moreover, the contribution of customers' decision-making is substantial to the sustainable progress of e-service management in this research area. Accordingly, they can direct the hospitality and tourism business to obtain a long-term competitive advantage through efficient management.

### *Limitations*

Our work has some limitations, such as unavailability and inaccessibility of data, time and budget constraints of the researchers, and some other practical concerns. The present study has only investigated the research scope in hotels from 4-star to 5-star. Therefore, the findings might not be transferable to below 3-star hotels. The most important limitation lies in the items of guest reviews online of a hotel. They can be measured by other direct and indirect factors, such as the internet speed, the interface of the web pages, the hotel's marketing mix, etc. The respondents may be confused with the information provided by the questionnaire.

Moreover, they may have different levels of knowledge and experience in completing the questionnaire. This factor might affect the outcomes of the designed framework. The collecting data area is from one region only as Ho Chi Minh City. The limit of time the researchers spend on asking respondents to do questionnaires on the offline approach might lead to inappropriate responses for data analysis.

### *Future studies*

There are other factors influencing service quality based on guest online reviews need for future study in the literature of various research areas. Future studies should target other perspectives on culture, economic issues, government policy, employee performance, salary, etc. The design of advanced internet applications and the development of the next-demanding up of customers will challenge us for years. In addition, the deeper application of customer engagement behaviors in different nationalities, backgrounds and so on might be interesting aspects to study. It means the further investigation of the broader world is in great need.

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